

Evidence + Investigation Layer

Updated Implementation Specification — Investigation-first, Evidence-backed, LLM deferred

1. Existing Platform — Do NOT Rebuild

- Data ingestion
- Data profiling
- Schema validation
- KPI detection
- KPI selection
- KPI definition
- SQL Server integration
- Parquet-based dataset storage
- DuckDB analytical access
- Anomaly Detection
- Dimension Analysis
- Contribution Analysis
- Hierarchical Drill-Down

Rule: inspect and reuse the existing implementations through clean interfaces. Do not create duplicates.

2. Capabilities Already Visible in the UI

Capability	Existing UI capability
KPI Change	Previous/current period, absolute and percentage change, measure, time column, comparison, period length, rows compared.
Contribution	Primary drivers, secondary drivers, and offsetting factors.
Segment status	NEW and GONE segment detection.
Hierarchy	Airline → Sentiment → Cabin, including concentration where a whole segment shares a value.
Evidence summary	Rows scanned, period rows, rows outside periods, reconciliation, query count and execution time.

3. Target Architecture — No LLM Yet

The current phase must be fully deterministic. LLM integration is explicitly deferred.

Investigation Request → Planner → Orchestrator → Existing Analysis Engines → Evidence Builder → Evidence Validator → Contributor/Root-Cause Ranking → Investigation Result → Evidence UI

4. Investigation Engine & Planner

Create an InvestigationEngine that orchestrates existing KPI, Dimension, Contribution, Drill-Down, Anomaly, and DuckDB capabilities. The planner must work with arbitrary KPIs, dimensions, periods, and datasets.

5. Investigation State & Lifecycle

Persist question, KPI, periods, findings, contributors, drill-down path, evidence IDs, validation, limitations, execution metadata, and status.

PLANNED → RUNNING → COMPLETED / PARTIAL / FAILED

6. Evidence Layer

Evidence is a backend concept first and a frontend representation second. Every important numerical or analytical finding needs structured evidence.

```
Evidence fields: id, claim, evidence_type, metric, dimension, dimension_value, previous_period, current_period,
previous_value, current_value, absolute_change, percentage_change, contribution_percentage, filters,
source_dataset, source_relation, source_columns, query, analysis_tool, validation_status, confidence
```

7. Evidence Types

Support: kpi_change, dimension_change, contribution, drill_down, anomaly, trend, comparison, validation, new_segment, gone_segment, offsetting_factor, execution, coverage, reconciliation.

8. Analysis Execution Evidence — NEW

Promote the metrics already shown in the Evidence UI into formal evidence:

```
{ "rows_scanned": 197221, "rows_in_previous_period": 22, "rows_in_current_period": 21, "rows_outside_periods":
197178, "queries_executed": 7, "execution_time_ms": 141, "contribution_reconciliation": "PASSED" }
```

9. Contribution Reconciliation — NEW

Validate that the complete contribution decomposition reconciles to the total KPI movement, using a configurable tolerance. The complete decomposition may reconcile to 100% even when displayed Primary/Secondary/Offsetting subsets do not sum to 100%.

10. Dimension Explainability Score — NEW

Keep “Explains” separate from Contribution. Contribution answers how much a segment contributed; Explainability measures how strongly segment movement deviates from proportional movement. Values above 100% are not automatically errors.

```
Contribution → how much did a segment contribute? Explainability → how differently did segments behave from
proportional movement?
```

11. NEW / GONE + Offsetting Factors

Treat NEW and GONE as first-class evidence. Keep Primary Drivers, Secondary Drivers, and Offsetting Factors as explicit analytical categories. Do not imply causality from contribution alone.

12. Investigation Decision Trace — NEW

For each important decision, store why the system selected a segment or dimension.

```
Selected: Singapore Airlines Why: contribution = 80%, rank = #1, absolute change = -12, status = GONE Drill-down
selected: Sentiment Why: relevant available dimension with meaningful concentration Drill-down stopped: Cabin
why: threshold reached / no useful dimensions / insufficient data / no additional explanatory power
```

13. Drill-Down Stop Reasons — NEW

Every drill-down must have a deterministic stop reason: maximum useful depth, remaining contribution below threshold, no suitable dimensions, insufficient data, or no additional explanatory power.

14. Evidence Quality & Validation — NEW

Add an explicit Evidence Quality summary separate from analytical confidence.

✓ Data/period coverage ✓ Numerical consistency ✓ Contribution reconciliation ✓ Query provenance ✓ Source traceability ✓ Required metadata

15. Query Trace — NEW

Turn “7 queries in 141 ms” into inspectable evidence. Add View Queries showing each actual query, purpose, execution time, rows returned, and status. Never fabricate SQL.

16. Investigation Audit Trail — NEW

Store events such as investigation started, KPI calculated, dimension analysis executed, contributor selected, drill-down executed, evidence validated, reconciliation passed, and investigation completed.

17. Investigation Tree / Evidence Graph

Link every important tree node to evidence IDs so the hierarchy becomes an evidence-backed investigation graph.

Value For Money -15 ■■■ Airline: Singapore Airlines -12 / 80% ■■■ Sentiment: positive -12 / 80% ■■■ Cabin: Economy -12 / 80%

18. Frontend Architecture — Recommended

Evidence should live inside the Investigation page, but as a visually separate section. Do not make it a separate top-level analysis page at this stage.

Investigation ■■■ Overview / KPI Change ■■■ Primary Drivers ■■■ Secondary Drivers ■■■ Offsetting Factors ■■■ Hierarchical Drill-Down ■■■ Evidence & Validation

Investigation answers “What did we find?”. Evidence answers “Why should I trust this analysis?”.

19. Evidence & Validation UI

Keep the current Evidence panel and make it collapsible. Show summary metrics first; expose detailed provenance and SQL through actions.

EVIDENCE & VALIDATION Evidence Quality: VALIDATED Rows scanned 197,221 Rows compared 43 Queries executed 7 Reconciliation ✓ 100% [Show details] [View queries] [View source]

20. API Requirements

POST /api/investigations GET /api/investigations/{id} GET /api/investigations/{id}/evidence GET /api/investigations/{id}/tree GET /api/investigations/{id}/audit GET /api/investigations/{id}/queries GET /api/evidence/{id}

21. Performance, Security & Reproducibility

Reuse cached analytical results, avoid duplicate queries, limit drill-down depth, record execution timing, and lazy-load detailed evidence. Respect existing authentication, RBAC, company/tenant isolation, and dataset ownership.

22. Testing Before LLM Phase

- KPI increase/decrease/zero and missing periods.
- Contribution reconciliation with tolerance.
- Primary/secondary/offsetting classification.
- NEW/GONE detection.
- Dimension Explainability, including values above 100%.
- Hierarchical drill-down and deterministic stop reasons.

- Evidence provenance and validation failures.
- Query trace correctness.
- Investigation lifecycle and audit trail.
- RBAC and tenant isolation.
- End-to-end investigation on the existing airline/sales dataset.

23. End-to-End Acceptance Test

Run an investigation such as “Why did Value For Money decrease?” using the existing dataset and engines.

Expected: Value For Money: 65 → 50 Change: -15 Percentage change: -23.1% Primary driver: Singapore Airlines – -12 – 80% – GONE Secondary: LOT Polish Airlines, Saudia, Air India, Korean Air, Emirates Offsetting: TAP Portugal, Turkish Airlines, Aer Lingus, etc. Hierarchy: Singapore Airlines → Sentiment: positive → Cabin: Economy Evidence: Rows scanned, period rows, reconciliation, queries, execution time, provenance, validation, decision trace, audit trail.

24. Phase Boundary — LLM Deferred

Do not implement LLM integration in this phase. The deterministic Investigation + Evidence system must be complete, validated, traceable, reproducible, and useful on its own. The future LLM phase will consume the validated Investigation Result.

25. Final Pre-LLM Checklist

- Investigation is deterministic and reproducible.
- Existing analytical engines are reused.
- Every important finding has structured evidence.
- Evidence has source/query provenance.
- Contribution decomposition reconciles.
- Dimension Explainability is separate from Contribution.
- NEW/GONE and offsetting factors are explicit.
- Drill-down decisions and stop reasons are recorded.
- Evidence Quality is validated.
- Query execution is traceable.
- Investigation status is explicit.
- Audit trail is available.
- Frontend shows findings first and Evidence & Validation as a separate section inside the Investigation page.
- RBAC and tenant isolation are enforced.
- End-to-end tests pass.

Key Principle

Investigation = what the system discovered.

Evidence = why the discovery is trustworthy.

Existing analytical engines = how the numbers are computed.

LLM = a later explanation layer, not part of this phase.